STAT 471: Introduction to R Programming Lecture

Lecture 14: Monte Carlo Markov Chains (MCMC) and Other Monte Carlo Techniques

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Overview

1. Introduction to Acceptance-Rejection Sampling
2. Introduction to Monte Carlo Markov Chains
3. MCMC Samplers: Metropolis-Hastings and Gibbs Samplers

Objectives

I'll try to minimize the amount of theory this time around since we covered a lot of theory for the bootstrap and jacknife resampling methods:

- Understand how to apply Bayesian statistical theory to generate samples from a target distribution
- Get to know how to carry out acceptance-rejection sampling, a core algorithm used in MCMC techniques
- Understand how MCMC can be applied to other settings

Acceptance-Rejection Algorithm

- We generate samples randomly and "accept" any samples that fall within our target distribution's probability density function
- Suppose that X and Y are random variables with pdf or pmf f and g respectively.
- Suppose that there is a constant c such that $\frac{f(t)}{g(t)} < c$

The Acceptance-Rejection Method

1. Find a random variable Y with density g satisfying $f(t)/g(t) \leq c$, for all t such that $f(t) > 0$. Provide a method to generate random Y.

2. For each random variate required:

   (a) Generate a random y from the distribution with density g.

   (b) Generate a random u from the Uniform(0, 1) distribution.

   (c) If $u < f(y)/(cg(y))$, accept y and deliver $x = y$; otherwise reject y and repeat from Step 2a.

Why the Acceptance-Rejection Algorithm Works

$$P(\text{accept} \mid Y) = P\left(U < \frac{f(Y)}{cg(Y)} \,\middle|\, Y\right) = \frac{f(Y)}{cg(Y)}$$

The total probability of acceptance for any iteration is:

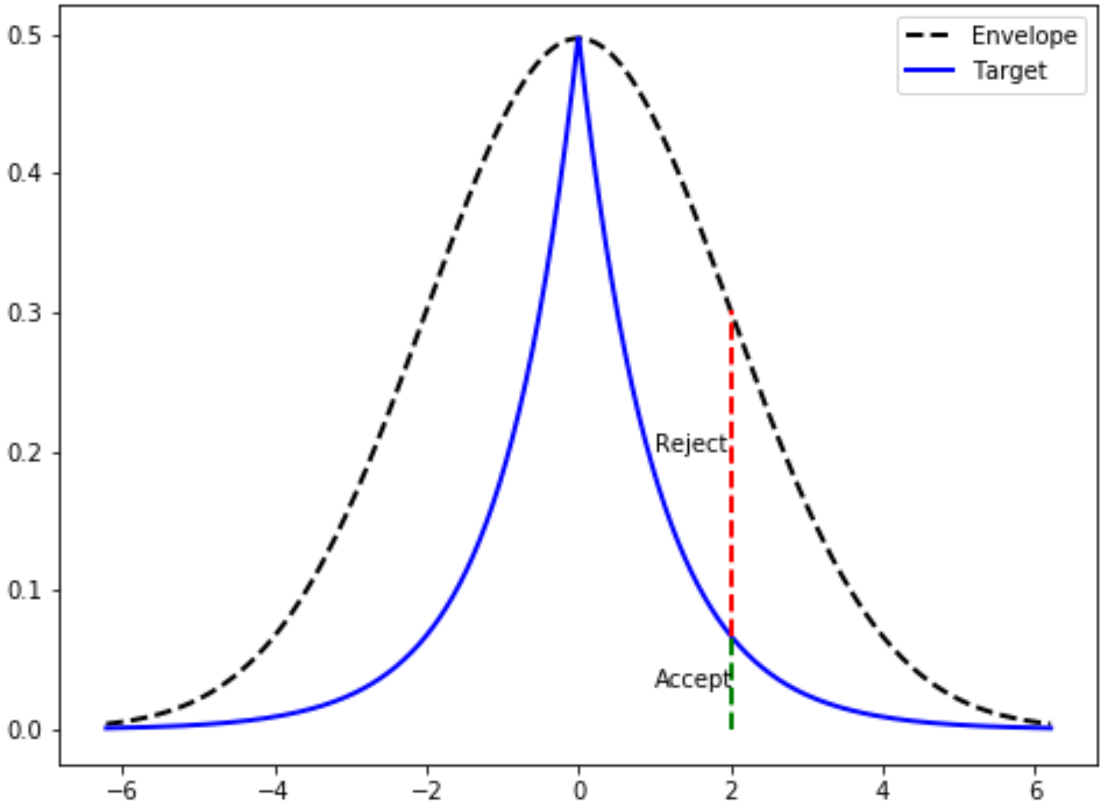
$$\sum_y P(\text{accept} \mid y) P(Y=y) = \sum_y \frac{f(y)}{cg(y)} g(y) = \frac{1}{c}$$

So,

$$P(k \mid \text{accepted}) = \frac{P(\text{accepted} \mid k) g(k)}{P(\text{accepted})} = \frac{\left[\frac{f(k)}{c\,\cancel{g(k)}}\right] \cancel{g(k)}}{1/c} = f(k).$$

Continuous case follows by similar reasoning.

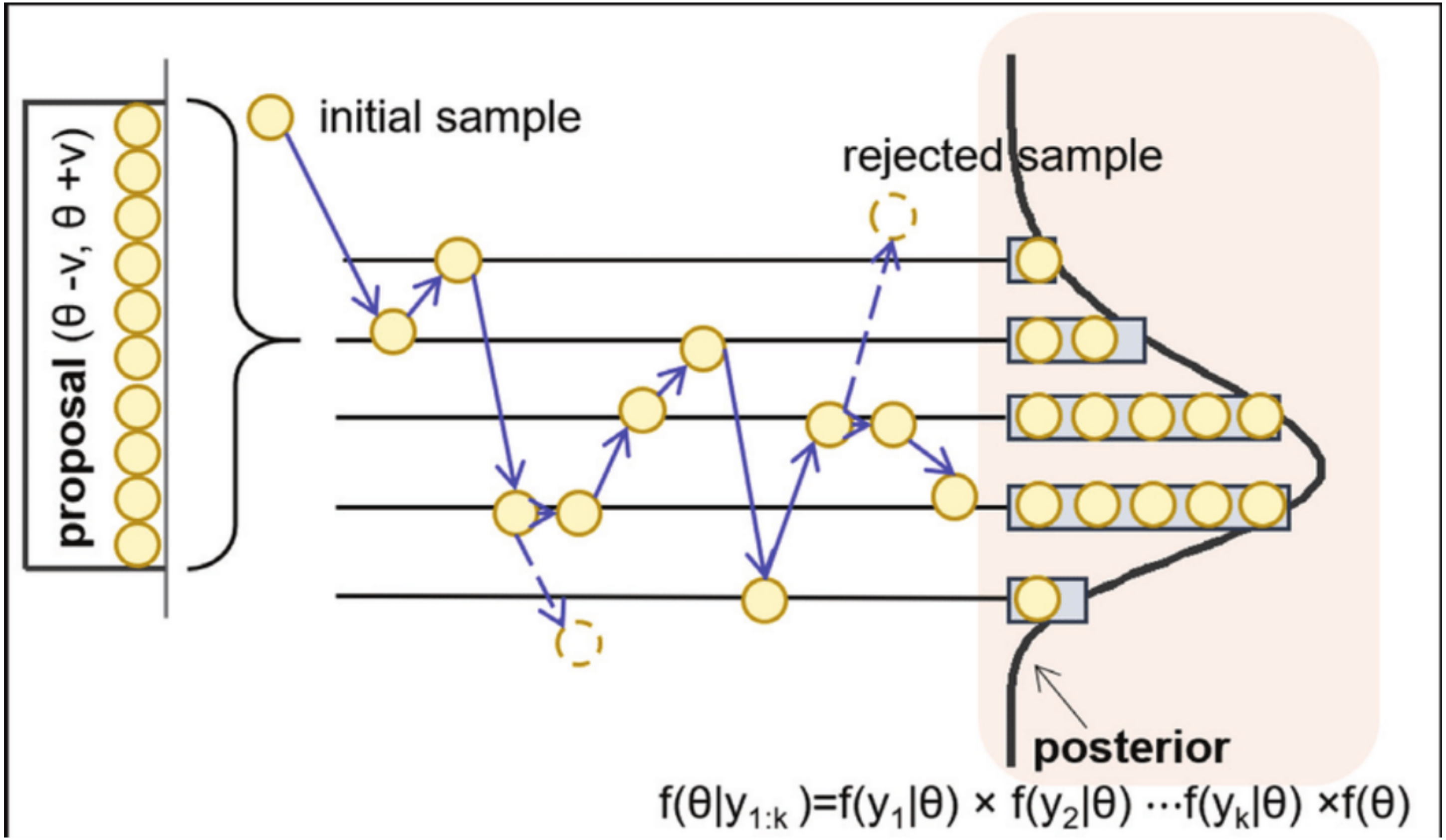
Example: Sampling from a Cauchy Distribution by Using a Normal Distribution via Acceptance-Rejection Sampling



Why Study MCMC?

- Monte Carlo sampling has been shown to be accurate for estimating probabilities and other hard problems such as integration.
- Idea: If Markov Chains can represent probability states of certain events happening, we can sample many Markov Chains and look at their stationary (where the chains start to converge tightly) distribution!
- Goal: Sample a large amount of Markov Chains to generate a distribution of Markov Chains that best represent our population parameters
- Applying Bayesian statistics: We treat our population as a posterior distribution, and we use acceptance-rejection sampling to sample from the candidate prior distribution.
- The MCMC Algorithm can be generally applied to the study of gene networks, epidemiology, risk modeling in finance, and generative models in Machine Learning

Visualization of MCMC



Metropolis-Hastings Algorithm

The *Metropolis-Hastings sampler* generates the Markov chain $\{X_0, X_1, \dots\}$ as follows.

1. Choose a proposal distribution $g(\cdot|X_t)$ (subject to regularity conditions stated above).
2. Generate X_0 from a distribution g.
3. Repeat (until the chain has converged to a stationary distribution according to some criterion):
   (a) Generate Y from $g(\cdot|X_t)$.
   (b) Generate U from Uniform(0,1).
   (c) If
   $$U \leq \frac{f(Y)g(X_t|Y)}{f(X_t)g(Y|X_t)}$$
   accept Y and set $X_{t+1} = Y$; otherwise set $X_{t+1} = X_t$.
   (d) Increment t.

Why M-H Sampling Works

Let $\alpha(X_t, Y) = \min\left(1, \frac{f(Y)g(X_t|Y)}{f(X_t)g(Y|X_t)}\right)$. Let r and s be two elements in the chain's state space. Suppose $f(s)g(r|s) \geq f(r)g(s|r)$. The joint density of (X_t, X_{t+1}) at (s,r) is:

$$f(s)g(r|s)\alpha(s,r) = f(s)g(r|s)\left(\frac{f(r)g(s|r)}{f(s)g(r|s)}\right) = f(r)g(s|r)\alpha(r,s)$$

Hence our chains converge in the state space to the stationary distribution f. $\square$

Gibbs Sampler Algorithm

Let $X = (X_1, \ldots, X_d)$ be a random vector in $\mathbb{R}^d$. Define the $d-1$ dimensional random vectors

$$X_{(-j)} = (X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_d),$$

and denote the corresponding univariate conditional density of X_j given $X_{(-j)}$ by $f(X_j|X_{(-j)})$. The Gibbs sampler generates the chain by sampling from each of the d conditional densities $f(X_j|X_{(-j)})$.

In the following algorithm for the Gibbs sampler, we denote X_t by $X(t)$.

1. Initialize $X(0)$ at time $t = 0$.
2. For each iteration, indexed $t = 1, 2, \ldots$ repeat:
   (a) Set $x_1 = X_1(t-1)$.
   (b) For each coordinate $j = 1, \ldots, d$
      i. Generate $X_j^*(t)$ from $f(X_j|x_{(-j)})$.
      ii. Update $x_j = X_j^*(t)$.
   (c) Set $X(t) = (X_1^*(t), \ldots, X_d^*(t))$ (every candidate is accepted).
   (d) Increment t.

Why Gibbs Sampling Works

Pf. Let T be the random variable generated by the Gibbs Sampler.

Then,

$$P(T \le t) = E[F_{X|Y}(t)] = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{t} f_{X|Y}(x|y)\,dx\right] f_Y(y)\,dy$$

$$= \int_{-\infty}^{t}\left[\int_{-\infty}^{\infty} f_{X|Y}(x|y)\, f_Y(y)\,dy\right]dx$$

$$= \int_{-\infty}^{t}\left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy\right]dx$$

$$= \int_{-\infty}^{t} f_X(x)\,dx$$

Hence the random variable generated by the algorithm has pdf $f_X(x)$. $\square$

Next Time...

Last Lecture: Introduction to Time Series Analysis

Announcements

Programming Assignment 4 (Last One) due this Saturday at 11:59pm